

Markdown-to-Lout Pipeline Status

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mdlout project

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1. Introduction

This report documents the current state of the **mdlout** Markdown-to-Lout converter. The toolchain is split into three layers: a single-file Python front end that parses Markdown and emits Lout source, a forked C implementation of Lout (3.43, William8000 fork) that turns Lout into PostScript or SVG, and a thin wrapper that produces PDF (via `ps2pdf`) or HTML (by wrapping the SVG back end's output).

1.1. Architecture

The front end runs four phases: frontmatter extraction, block-level parsing, inline conversion via a placeholder system that protects code and links from double-escaping, and Lout generation. For `report` and `book` document types, heading levels are mapped to nested `@Section / @SubSection` constructs; for the default `doc` type, headings render as styled `@Display` blocks.

1.1.1. Why Lout?

Lout's stream model $\hat{a}, \sim$ galleys flowing into targets $\hat{a}, \sim$ is uniquely well-suited to mixing engraved music, mathematics, and prose without resorting to a TeX-style box-and-glue model. The trade-off is a smaller community and a narrower set of available macros, which is exactly the gap mdlout is filling.

2. Implementation notes

A representative slice of the inline conversion code:

```
def convert_inline(text: str) -> str:
    _ph_reset()
    result = _convert_inline_inner(text)
    return _ph_restore(result)
```

The placeholder dance keeps Lout's @ and { } metacharacters out of harm's way while inline spans are extracted, then restores them once plain-text escaping is finished.

2.1. A worked equation

The closed form for the n-th Fibonacci number, useful as a sanity check that math blocks survive the round-trip:

$$F_n = \frac{1}{\sqrt{5}} \left(\varphi^n - \psi^n \right), \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}$$

2.2. A figure block

Figure 1. The pipeline: Markdown @Arrow Lout @Arrow PostScript @Arrow PDF

3. Status and next steps

Working today: text emission, font lookup, page geometry, transforms, and underlining. In progress: the colour pipeline, raw SVG passthrough, and the three macro shims @Math, @ABC, @SVG. Once those land, mdlout will be able to typeset prose, math, and engraved music in one document â,~ which is the original goal.